

REQUIREMENT ANALYSIS DOCUMENT

StockSense

AI-Powered Inventory Shortage Predictor for Small Businesses

(Tentative Title)

Domain: Artificial Intelligence & Machine Learning with Web Development

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Project Guide

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1. Introduction

1.1 Purpose

This Requirement Analysis Document (RAD) defines the functional and non-functional requirements for StockSense, an AI-powered web application designed to help small businesses predict inventory shortages before they occur. It serves as a reference for the development team, project guide, and evaluators to understand the scope, features, and constraints of the system.

1.2 Project Scope

StockSense will collect historical sales and stock-level data from a small business, process it, and apply a machine learning model to forecast upcoming inventory shortages. Predictions are surfaced through a web dashboard with proactive alerts, enabling business owners to reorder stock in time and avoid lost sales. The system targets small and medium retail businesses that currently rely on manual or spreadsheet-based inventory tracking.

1.3 Intended Audience

- Project Guide and evaluation panel, for reviewing project direction and progress.
- Development team (project members), as the primary technical reference.
- Prospective end users (small business owners), to validate that requirements reflect real needs.

2. Overall Description

2.1 Product Perspective

StockSense is a new, standalone web-based system. It is not a modification of an existing product. It consists of a machine learning prediction engine, a backend service layer, and a web-based front-end dashboard, all working together to convert raw sales and stock data into actionable shortage predictions.

2.2 Product Features (Summary)

- Upload and management of sales and stock data.
- ML-based prediction of inventory shortages per product.
- Automated alerts and notifications before stock runs out.
- Web dashboard with visual insights into stock trends.
- Basic user authentication and role-based access (owner / admin).

2.3 User Classes and Characteristics

User Class	Description	Technical Expertise
Business Owner	Primary user; views dashboard, receives shortage alerts, manages products.	Low to Moderate
System Administrator	Manages user accounts and oversees model retraining.	Moderate to High

2.4 Operating Environment

StockSense operates as a web application accessible via any modern browser (Chrome, Firefox, Edge) on desktop or mobile devices. The backend and ML model are hosted on a server environment supporting Python-based services and a relational/NoSQL database.

2.5 Assumptions and Dependencies

- Business owners will provide reasonably consistent historical sales and stock data.
- A stable internet connection is available for accessing the web dashboard.
- Sufficient historical data (minimum a few months) is required for meaningful predictions.

3. Functional Requirements

The functional requirements are grouped by module. Priority is rated as High (H), Medium (M), or Low (L).

3.1 User & Access Management

ID	Requirement	Description	Priority
FR-01	User Registration & Login	The system shall allow business owners and admins to register and log in securely.	H
FR-02	Role-Based Access	The system shall restrict features based on user role (Owner / Admin).	M

3.2 Data Management

ID	Requirement	Description	Priority
FR-03	Upload Sales Data	The system shall allow users to upload or sync sales transaction records.	H
FR-04	Upload Stock Data	The system shall allow users to input and update current stock levels.	H
FR-05	Manage Product Catalog	The system shall allow users to add, edit, or remove products.	M

3.3 Prediction & Alerts

ID	Requirement	Description	Priority
FR-06	Shortage Prediction	The system shall use a trained ML model to predict likely shortage dates per product.	H
FR-07	Automated Alerts	The system shall generate and send an alert when predicted stock is nearing a shortage threshold.	H
FR-08	Configurable Thresholds	The system shall allow users to set custom reorder threshold levels.	M
FR-09	Model Retraining	The system shall support periodic retraining of the prediction model with new data.	L

3.4 Dashboard & Reporting

ID	Requirement	Description	Priority
FR-10	Inventory Dashboard	The system shall display current stock status and shortage risk in a visual dashboard.	H
FR-11	Trend Visualization	The system shall visualize historical sales and stock trends via charts.	M
FR-12	Report Generation	The system shall allow users to generate/export basic inventory reports.	L

4. Non-Functional Requirements

Category	Requirement
Performance	The system shall generate shortage predictions within a few seconds for a typical small-business catalog.
Usability	The dashboard shall be simple and intuitive enough for users with minimal technical expertise.
Reliability	The system shall maintain consistent alert delivery with minimal downtime.
Security	User data and credentials shall be stored securely, with authenticated access to all features.
Scalability	The system architecture shall support a growing number of products and users over time.
Maintainability	The codebase shall be modular to allow easy updates to the ML model or web interface.
Portability	The web dashboard shall be accessible across major browsers and screen sizes.

5. External Interface Requirements

5.1 User Interfaces

A responsive web dashboard providing login, data upload, product management, prediction views, and alert notifications.

5.2 Hardware Interfaces

No specialized hardware is required. The system runs on standard server infrastructure and is accessed via standard client devices (desktop/laptop/mobile).

5.3 Software Interfaces

- Backend: Python-based REST API framework.
- Machine Learning: Scikit-learn / Pandas / NumPy for model training and inference.
- Database: Relational or NoSQL database for storing sales, stock, and user data.

- Frontend: Web framework (HTML/CSS/JavaScript-based) for the dashboard.

5.4 Communication Interfaces

Communication between the frontend, backend, and ML service occurs over HTTPS using RESTful API calls.

6. System Requirements

Component	Minimum Requirement
Server (Development)	Quad-core CPU, 8 GB RAM, 20 GB storage
Client Device	Any modern device with a web browser and internet access
Operating System (Server)	Linux / Windows Server
Programming Language	Python (ML & backend), JavaScript/HTML/CSS (frontend)
Database	PostgreSQL / MySQL / MongoDB (as finalized by the team)

7. Constraints and Assumptions

- The project must be completed within the academic semester timeline.
- Prediction accuracy depends on the quality and volume of historical data provided.
- The initial version targets single-outlet small businesses; multi-branch support is a future scope item.
- The team has limited computing resources, favoring lightweight, efficient ML models over deep learning where possible.

8. Glossary

Term	Definition
Shortage	A situation where stock of a product falls to zero or below demand before it is replenished.
ML Model	A machine learning algorithm trained on historical data to make predictions.
Dashboard	The web interface presenting stock status, alerts, and trends to the user.
Reorder Threshold	A configurable stock level that triggers a shortage alert.